PHY 321 1.0 Medical Physics

Field Visit Report

Apeksha Hospital – Maharagama

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Introduction

Apeksha Hospital is the main government cancer treatment hospital in Maharagama, Sri Lanka. It is the largest specialized cancer hospital in the country and provides free treatment for thousands of patients every year. The hospital offers many services including cancer screening and diagnosis, chemotherapy, radiotherapy, surgeries, pediatric oncology for children, and palliative care for advanced patients. It has experienced doctors, nurses, and modern medical equipment to treat different types of cancer. Patients from all over Sri Lanka come to this hospital because it is a leading center for cancer care and gives hope and support to patients and their families.

During our field visit to Apeksha Hospital, we studied nine important units one by one to understand how cancer treatment is carried out. These units included **the Water Phantom unit, Mechanical unit, Daily QA (Quality Assurance), LINAC Planning, Patient-Specific QA, CT Simulation and Brachytherapy, Radiation Protection, PET unit, and the Iodine and Cobalt treatment units**. Through this visit, we learned how each unit plays an important role in radiation therapy, treatment planning, quality checking, patient safety, and accurate cancer diagnosis and treatment. This field visit helped us gain practical knowledge about modern cancer treatment technologies and safety procedures used in the hospital.

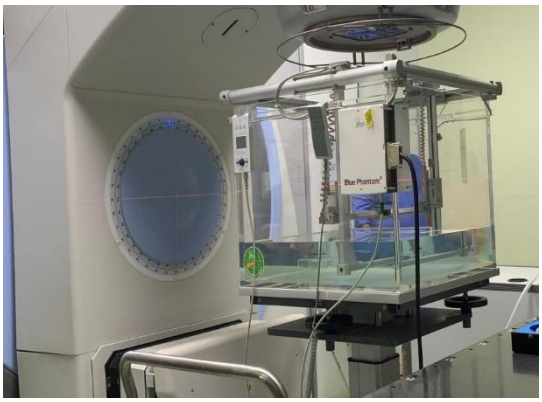
Now we will consider each units and study how they are used with Physics.

1.The Water Phantom unit

The Water Phantom Unit at Apeksha Hospital is mainly used for radiation dose measurement and calibration in radiotherapy. This unit ensures that the exact amount of radiation planned for a patient is accurately delivered. A water phantom is a large water-filled tank that acts like the human body because water has similar radiation absorption properties to human tissue.

The main machines used in this unit include the water phantom tank, a 3D scanning system, ionization chambers, detectors, an electrometer, and a computer system for data analysis. The ionization chamber is placed inside the water tank to measure radiation dose at different depths and positions. The scanning system moves the detector automatically in different directions (X, Y, Z) to record precise dose distribution.

The main activities in this unit include beam calibration, dose measurement, beam profile measurement, percentage depth dose (PDD) measurement, and quality assurance testing of the LINAC machine. These tests are very important to maintain accuracy, patient safety, and treatment quality in radiation therapy.



2. Mechanical unit

The Mechanical Unit in the radiotherapy department is responsible for checking and maintaining the mechanical accuracy of radiation treatment machines such as the LINAC (Linear Accelerator). This unit ensures that all moving parts of the machine work properly and safely. The main purpose is to confirm that the machine delivers radiation exactly to the planned position without any mechanical errors.

The main equipment used in this unit includes mechanical QA tools such as front pointers, graph papers, spirit levels, laser alignment systems, gantry angle indicators, collimator rotation checks, and treatment couch position measurement devices. These tools help to check the alignment of lasers, accuracy of gantry rotation, collimator angles, field size accuracy, and patient couch movement.

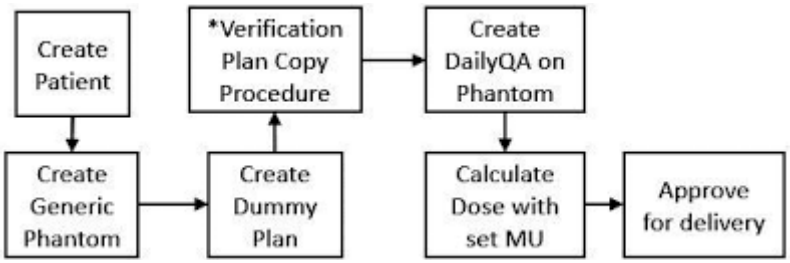
The main activities of the Mechanical Unit include daily, monthly, and annual mechanical quality assurance (QA) tests. Technicians and medical physicists carefully measure and record machine parameters to ensure proper alignment and positioning. This unit plays a very important role in maintaining treatment accuracy, patient safety, and the overall quality of radiation therapy.

3. Daily QA (Quality Assurance) unit

The Daily QA (Quality Assurance) Unit is responsible for checking the performance and safety of radiation therapy machines every day before patient treatment begins. The main purpose of this unit is to ensure that the LINAC (Linear Accelerator) delivers the correct radiation dose accurately and safely. Daily QA is very important because even small errors in radiation dose or machine alignment can affect patient treatment.

The main equipment used in this unit includes QA phantoms, ionization chambers, electrometers, detector arrays, and laser alignment tools. These devices are used to measure output constancy, beam symmetry, flatness, laser accuracy, and mechanical alignment. Medical physicists and radiation technologists perform these tests each morning and compare the results with standard reference values.

The main activities of the Daily QA Unit include checking radiation output consistency, verifying beam energy, testing safety interlocks, and confirming proper machine functioning. If any parameter is outside the acceptable limit, the machine is not used until the problem is corrected. This unit plays a critical role in maintaining treatment accuracy, machine reliability, and patient safety in radiotherapy.



4. LINAC Planning unit

The LINAC Planning Unit is responsible for planning radiation treatment for cancer patients using a Linear Accelerator (LINAC). The main purpose of this unit is to design an accurate treatment plan that delivers the correct radiation dose to the tumor while protecting healthy tissues. This planning process is very important to ensure effective and safe cancer treatment.

The main equipment used in this unit includes the Treatment Planning System (TPS) computer software, CT simulation images, contouring tools, and dose calculation systems. First, CT images of the patient are taken from the CT Simulation unit. Then, radiation oncologists and medical physicists carefully outline the tumor and nearby organs on the computer. After that, they calculate the radiation dose, beam angles, and treatment technique such as IMRT or 3D conformal therapy.

The main activities in this unit include image importing, target contouring, dose calculation, plan optimization, and plan evaluation. Once the treatment plan is approved, it is transferred to the LINAC machine for patient treatment. The LINAC Planning Unit plays a vital role in ensuring precise, safe, and effective radiation therapy for cancer patients.

5. Patient-Specific QA unit

The Patient-Specific Quality Assurance (QA) Unit is responsible for verifying each individual patient's radiation treatment plan before actual treatment begins. The main purpose of this unit is to ensure that the radiation dose calculated in the Treatment Planning System (TPS) is accurately delivered by the LINAC machine. This step is especially important for advanced treatment techniques such as IMRT and VMAT.

The main equipment used in this unit includes QA phantoms (solid water or detector phantoms), ionization chambers, 2D/3D detector arrays, electrometers, and analysis software. The patient's treatment plan is delivered to a phantom instead of the patient. The detectors inside the phantom measure the actual radiation dose, and the results are compared with the planned dose calculated by the computer system.

Physics plays a very important role in this unit. Medical physics principles such as radiation interaction with matter, dose measurement, beam attenuation, scattering, and inverse square law are applied. Physicists use dosimetry calculations to measure absorbed dose accurately. They also apply concepts like percentage depth dose (PDD), beam profiles, and gamma index analysis to compare measured and calculated dose distributions. If the measured dose matches the planned dose within acceptable limits, the treatment is approved.

This unit ensures treatment accuracy, patient safety, and high-quality radiation therapy by applying practical medical physics knowledge.

6. CT Simulation and Brachytherapy unit

The CT Simulation and Brachytherapy Unit is a critical part of radiation therapy, combining imaging and treatment techniques to ensure precise targeting of tumors.

CT Simulation is used to create detailed 3D images of a patient's anatomy. These images allow radiation oncologists and medical physicists to locate the tumor accurately and define the treatment area. The patient lies on a flat table, and a CT scanner takes cross-sectional images. These images are then used in the Treatment Planning System (TPS) to design the radiation plan. Physics principles such as X-ray attenuation, Hounsfield units, and geometric accuracy are used to convert CT image data into accurate density information for dose calculations.

Brachytherapy is a treatment method where radioactive sources are placed directly inside or very close to the tumor. This allows delivery of high radiation doses to the tumor while sparing surrounding healthy tissue. The unit uses applicators, radioactive seeds, and specialized planning software. Physics concepts such as inverse square law, dose rate, decay of radioactive sources, and tissue absorption are applied to calculate the correct dose and placement of sources.

The main activities of this unit include patient positioning, CT imaging, contouring the tumor and organs at risk, planning the brachytherapy dose, and performing treatment with careful verification. By combining CT simulation and brachytherapy, the unit ensures precise, safe, and effective treatment, applying medical physics principles at every step.

7.Radiation Protection

The Radiation Protection Unit is responsible for ensuring the safety of patients, staff, and the public from unnecessary exposure to radiation. Its main role is to implement safety measures, monitor radiation levels, and maintain protocols so that all radiation used in diagnosis and treatment is controlled and safe.

The main equipment used in this unit includes radiation detectors, Geiger-Müller counters, dosimeters, survey meters, shielding materials, and monitoring software. Staff wear personal dosimeters to track cumulative radiation exposure, and survey meters are used to check radiation levels in treatment rooms, storage areas, and surrounding zones. Shielding walls, lead aprons, and barriers are used to reduce exposure in clinical areas.

Physics is fundamental to this unit. Concepts such as the inverse square law, attenuation, half-value layer, scatter radiation, and radiation dose units (Gray and Sievert) are applied to calculate safe distances, shielding thickness, and exposure limits. Dosimetry calculations allow physicists to ensure that doses remain within safe limits, and the behavior of radiation in matter is used to design protective measures.

The main activities of this unit include radiation monitoring, staff training, room shielding verification, radiation audits, and emergency response planning. By applying physics principles and strict safety protocols, the Radiation Protection Unit ensures that radiation is used effectively for treatment while minimizing risk to everyone.

8. PET (Positron Emission Tomography) unit

The PET (Positron Emission Tomography) Unit is a diagnostic imaging unit used to detect cancer, monitor treatment, and evaluate the spread of tumors in the body. It provides detailed information about metabolic activity in tissues, which helps doctors plan effective treatment.

The main equipment used in this unit includes a PET scanner, radiotracers (like FDG), gamma detectors, and computer systems for image reconstruction. A patient is injected with a small amount of radioactive tracer, which accumulates in active cancer cells. The PET scanner detects the gamma rays emitted when positrons from the tracer annihilate with electrons in the body. These signals are processed by the computer to create 3D images showing areas of high metabolic activity.

Physics plays a crucial role in the PET unit. Principles of positron decay, annihilation radiation, gamma photon detection, coincidence counting, and tomographic image reconstruction are used. Medical physicists apply knowledge of radiation interaction with matter, attenuation correction, and detector efficiency to ensure accurate imaging and minimize radiation dose to the patient.

The main activities in the PET unit include preparing radiotracers, positioning patients, scanning, processing images, and interpreting results. By applying physics principles, the PET Unit provides precise, non-invasive diagnostic information that is essential for cancer detection, staging, and treatment monitoring.

9. The Iodine and Cobalt treatment unit

The Iodine and Cobalt Treatment Unit is a specialized radiotherapy unit used to treat cancer using radioactive isotopes. Iodine-131 is primarily used for treating thyroid cancer, while Cobalt-60 is used for external beam radiotherapy for various cancers. This unit ensures targeted delivery of radiation to tumors while minimizing exposure to healthy tissues.

The main equipment in this unit includes Cobalt-60 teletherapy machines, sealed radioactive Iodine sources, patient applicators, shielding devices, and treatment planning systems. For Iodine therapy, patients may ingest or be injected with radioactive iodine, which is absorbed by thyroid tissue. In Cobalt therapy, a Cobalt-60 source emits gamma rays that are directed toward the tumor using a treatment machine, while protective shielding and precise beam shaping minimize dose to surrounding healthy tissue.

Physics is critical in this unit. Concepts such as radioactive decay, half-life, gamma radiation, inverse square law, dose calculation, tissue absorption, and shielding design are applied to determine the correct activity, treatment duration, and safe distance. Medical physicists calculate the required radiation dose for effective tumor control and ensure that exposure to staff and surrounding areas is minimized.

The main activities of this unit include treatment planning, dose calculation, patient positioning, source calibration, radiation monitoring, and safety checks. By applying medical physics principles, the Iodine and Cobalt Treatment Unit delivers precise, effective, and safe radiotherapy to cancer patients.

Conclusion

Our field visit to Apeksha Hospital for the Medical Physics course gave us valuable practical knowledge about cancer treatment and radiotherapy. We learned how different units, from Water Phantom and Mechanical to LINAC Planning, Patient-Specific QA, CT Simulation, Brachytherapy, Radiation Protection, PET, and Iodine & Cobalt treatment, work together to deliver safe and precise radiation therapy. The visit helped us understand the important role of medical physics in treatment planning, dose measurement, quality assurance, and patient safety. Overall, this experience enhanced our understanding of how physics is applied in real clinical settings and highlighted the importance of accuracy, safety, and teamwork in cancer treatment.